

## LRU Pseudo Code:

```
n = int(input("Enter number of pages: "))
pages = list(map(int, input("Enter page reference string: ").split()))
f = int(input("Enter number of frames: "))

frames = [-1] * f
order = []

hit = 0
miss = 0

for i in range(n):
    page = pages[i]

    if page in frames:
        hit += 1
        print(f"Page {page} -> HIT -> ", end="")

        # update recent
        if page in order:
            order.remove(page)
        order.append(page)

    else:
        miss += 1
        print(f"Page {page} -> FAULT -> ", end="")

        if -1 in frames:
            idx = frames.index(-1)
            frames[idx] = page
        else:
            # remove least recent
            lru_page = order.pop(0)
            idx = frames.index(lru_page)
            frames[idx] = page

        order.append(page)

    # print status
    for j in range(f):
        print(frames[j], end=" ")
    print()

print("\nPage Miss =", miss)
print("Page Hits =", hit)
```

Input/Output:

Number of Pages: 10

Page String: 7 0 1 2 0 3 0 4 2 3

Frame Size: 3

```
... Page 7 -> FAULT -> 7 -1 -1
Page 0 -> FAULT -> 7 0 -1
Page 1 -> FAULT -> 7 0 1
Page 2 -> FAULT -> 2 0 1
Page 0 -> HIT -> 2 0 1
Page 3 -> FAULT -> 2 0 3
Page 0 -> HIT -> 2 0 3
Page 4 -> FAULT -> 4 0 3
Page 2 -> FAULT -> 4 0 2
Page 3 -> FAULT -> 4 3 2
```

Page Miss = 8

Page Hits = 2

Optimal Pseudo Code:

```
n = int(input("Enter number of pages: "))
pages = list(map(int, input("Enter page reference string: ").split()))
f = int(input("Enter number of frames: "))

frames = [-1] * f

hit = 0
miss = 0

for i in range(n):
    page = pages[i]

    if page in frames:
        hit += 1
        print(f"Page {page} -> HIT    -> ", end="")

    else:
        miss += 1
        print(f"Page {page} -> FAULT -> ", end="")

        if -1 in frames:
            idx = frames.index(-1)
        else:
            farthest = -1
            idx = -1

            for j in range(f):
                if frames[j] not in pages[i+1:]:
                    idx = j
                    break
                else:
                    next_use = pages[i+1:].index(frames[j])
                    if next_use > farthest:
                        farthest = next_use
            idx = j

        frames[idx] = page

    # print frame status
    for j in range(f):
        print(frames[j], end=" ")
    print()

print("\nPage Miss =", miss)
print("Page Hits =", hit)
```

Input/Output:

Number of Pages: 10

Page String: 7 0 1 2 0 3 0 4 2 3

Frame Size: 3

```
... Page 7 -> FAULT -> 7 -1 -1
    Page 0 -> FAULT -> 7 0 -1
    Page 1 -> FAULT -> 7 0 1
    Page 2 -> FAULT -> 2 0 1
    Page 0 -> HIT -> 2 0 1
    Page 3 -> FAULT -> 2 0 3
    Page 0 -> HIT -> 2 0 3
    Page 4 -> FAULT -> 2 4 3
    Page 2 -> HIT -> 2 4 3
    Page 3 -> HIT -> 2 4 3
```

Page Miss = 6

Page Hits = 4